Monte Carlo Pricing of a European Call: Milstein Discretisation under Geometric Brownian Motion

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Abstract

We specialise the Milstein scheme of the previous theoretical writeup to geometric Brownian motion, derive the resulting recursion, and apply it to Monte Carlo pricing of the European call. The principal purpose of the algorithm is not improvement of the pricer itself — Milstein and Euler share weak order 1 and produce statistically indistinguishable European prices — but rather *empirical demonstration of the strong/weak convergence asymmetry.* Running both schemes on the same grid of step sizes and on the same Brownian paths, we observe in a single experiment that Milstein achieves strong order 1 where Euler achieves only $\frac{1}{2}$, while the two are tied at weak order 1. This is the most direct possible confirmation of the theoretical result, and it is the experimental heart of the SDE discretisation notes.

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1 Introduction

The note on the Euler–Maruyama pricer implements Euler-Maruyama and verifies empirically that its strong order is $\frac{1}{2}$ and its weak order is 1. The asymmetry is a striking prediction of the SDE convergence theory developed in the note on SDE discretisation: the same scheme, applied to the same dynamics, has different convergence rates depending on which kind of error one measures.

The natural follow-up question is whether higher-order schemes lift both rates simultaneously, or only one. The answer, predicted by the theory, is that the Milstein scheme lifts the strong order from $\frac{1}{2}$ to 1 but leaves the weak order unchanged at 1. The present writeup verifies this prediction empirically on the same benchmark (European call pricing under GBM with all its inputs at canonical values) used for the Euler–Maruyama pricer, with the additional design choice that the Euler and Milstein experiments share their step-size grid so that the two slopes can be plotted on a single log-log axis.

The pricing application of the algorithm is, frankly, a sideshow. Both schemes give weak order 1 for European pricing under GBM, and the closed-form exact GBM sampler is unbiased and faster than either; Milstein adds nothing to the pricing pipeline that Euler did not already provide. The value of the algorithm is in the convergence study, where the asymmetry is demonstrated in code rather than asserted by a theorem. The practical relevance of Milstein in this project lies in pathwise sensitivity (Monte Carlo Greeks), where the strong order matters because the simulated path must be a faithful pathwise approximation of the true process for path-derivative estimators to make sense.

2 Milstein for geometric Brownian motion

2.1 The recursion

Recall the GBM SDE under the risk-neutral measure $\mathbb{Q}$:

$$\mathrm{d}S_t \;=\; rS_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t, \qquad S_0 > 0, \tag{1}$$

so $a(s) = rs$ and $b(s) = \sigma s$, hence $b'(s) = \sigma$ and $b(s)b'(s) = \sigma^2 s$. The Milstein recursion of the previous writeup specialises to

$$\hat{S}_{n+1} \;=\; \hat{S}_n + r\hat{S}_n h + \sigma\hat{S}_n\Delta W_n + \tfrac{1}{2}\sigma^2\hat{S}_n\big[(\Delta W_n)^2 - h\big], \qquad \hat{S}_0 = S_0, \tag{2}$$

with $\Delta W_n \sim \mathcal{N}(0,h)$ independent across n and $h = T/N$. Factoring $\hat{S}_n$ out:

$$\hat{S}_{n+1} \;=\; \hat{S}_n \cdot \Big(1 + rh + \sigma\Delta W_n + \tfrac{1}{2}\sigma^2\big[(\Delta W_n)^2 - h\big]\Big). \tag{3}$$

The recursion remains multiplicative, but the multiplicative factor is now *quadratic* in ΔW_n rather than linear as in Euler. This has two practical consequences worth noting.

2.2 Improved positivity

The Euler factor $1 + rh + \sigma\Delta W_n$ is unbounded below along the negative direction: a sufficiently negative ΔW_n produces a negative factor and hence a negative $\hat{S}_{n+1}$. The Milstein factor adds the term $\frac{1}{2}\sigma^2((\Delta W_n)^2 - h)$, which is bounded below by $-\sigma^2 h/2$: the $(\Delta W_n)^2$ component is always nonnegative, and only the $-\sigma^2 h/2$ piece pulls the factor down. Completing the square shows more:

$$1 + rh + \sigma\Delta W_n + \tfrac{1}{2}\sigma^2\big[(\Delta W_n)^2 - h\big] \;=\; \tfrac{1}{2}\big(1 + \sigma\Delta W_n\big)^2 + \tfrac{1}{2} + rh - \tfrac{1}{2}\sigma^2 h, \tag{4}$$

so the Milstein factor is strictly positive for every value of ΔW_n whenever $\sigma^2 h < 1 + 2rh$, which for $\sigma = 0.2$ and $r = 0.05$ holds for every h. On average, however, the Milstein factor is not closer to 1 than the Euler factor: both have mean $1 + rh$, and the variance of the Milstein factor exceeds that of the Euler factor by exactly $\sigma^4 h^2/2$, the variance of the corrector ($\mathrm{Var}((\Delta W_n)^2) = 2h^2$), which is uncorrelated with ΔW_n because $\mathbb{E}[(\Delta W_n)^3] = 0$.

We do not implement positivity defences in the algorithm, and none is needed: the Milstein factor is positive under the condition above, and the Euler factor is nonpositive with probability $\Phi(-(1+rh)/(\sigma\sqrt{h}))$ per step, about 3.7×10^{-57} at $h = 0.1$ and 3×10^{-546} at $h = 0.01$ (Section 2.2 of the Euler–Maruyama note).

2.3 The corrector term and its expectation

The new term $\frac{1}{2}\sigma^2((\Delta W_n)^2 - h)$ is the Milstein corrector. Two of its features deserve special note.

It has zero mean. Since $(\Delta W_n)^2$ has mean $\mathrm{Var}(\Delta W_n) = h$, the corrector has expectation zero: $\mathbb{E}[\frac{1}{2}\sigma^2((\Delta W_n)^2 - h)] = 0$. This is the mathematical reason that Milstein and Euler tie at weak order 1: the corrector affects each path differently, but its average contribution to $\mathbb{E}[f(\hat{S}_T)]$ is, to leading order, zero.

It is pathwise nonzero. Although the corrector vanishes on average, on any individual path $(\Delta W_n)^2 - h$ takes a specific value, typically of order h in absolute terms. This is the term that, in Euler, was discarded and contributed $O(h)$ to the strong error per step. In Milstein, it is included, and the next omitted term is of higher order: the strong error becomes $O(h)$ globally instead of $O(\sqrt{h})$.

The two features together explain the asymmetry. Strong convergence sees the pathwise contribution of the corrector and benefits from its inclusion. Weak convergence sees only the average contribution, which is zero either way. This is the conceptual content of the strong/weak distinction made concrete on a specific recursion.

2.4 Convergence orders

By Theorem 6.1 of the previous writeup applied to GBM,

$$\mathbb{E}[|S_T - \hat{S}_T^{\text{Milstein}}|] \;=\; O(h), \qquad |\mathbb{E}[f(S_T)] - \mathbb{E}[f(\hat{S}_T^{\text{Milstein}})]| \;=\; O(h) \quad \text{for } f \text{ smooth enough.} \tag{5}$$

Compare with Euler:

$$\mathbb{E}[|S_T - \hat{S}_T^{\text{Euler}}|] \;=\; O(h^{1/2}), \qquad |\mathbb{E}[f(S_T)] - \mathbb{E}[f(\hat{S}_T^{\text{Euler}})]| \;=\; O(h). \tag{6}$$

Strong order: $\frac{1}{2}$ (Euler) vs 1 (Milstein) — a genuine factor-$\sqrt{h}$ improvement at small h. Weak order: 1 (Euler) vs 1 (Milstein) — no improvement.

3 The biased estimator and its comparison with Euler

3.1 Definition

Let $\hat{S}_T^{(i),\,\text{Milstein}}$ for $i = 1, \ldots, M$ denote independent realisations of the Milstein-discretised terminal price. The Monte Carlo Milstein estimator of the European call price is

$$\hat{C}_{M,h}^{\text{Milstein}} \;\overset{\text{def}}{=}\; \frac{1}{M}\sum_{i=1}^{M} e^{-rT}\,(\hat{S}_T^{(i),\,\text{Milstein}} - K)^+. \tag{7}$$

The bias-variance decomposition of the Euler–Maruyama note, equation (10), applies here verbatim: total mean square error decomposes as variance/M plus bias-squared, with bias $O(h)$.

3.2 Comparison with Euler at fixed budget

Both schemes have weak order 1, so both produce a bias of order h for the same h. The constants in front of the h asymptote differ, and not in Milstein's favour: in Test 2 of the two validation scripts (common random numbers, $M = 2 \times 10^5$), the weak error at $N = 10$ is 4.5×10^{-2} for Milstein against 1.4×10^{-2} for Euler, and at $N = 100$ it is 4.5×10^{-3} against 1.1×10^{-3}. For this payoff the Milstein constant is three to four times larger.

The cost per path of Milstein is also higher: one extra multiplication and one squaring per Euler step (the corrector $(\Delta W_n)^2 - h$). In the benchmark tables of `python/results/phase3` (`pricing_*.csv`, 20 steps, one timing per case) the Milstein pricer takes 11–22% longer than the Euler pricer. At fixed total budget, therefore, Euler is the more accurate of the two for this payoff. At the sample sizes used for pricing, both biases are far below the statistical error (the half-width at $M = 20{,}000$ is about 0.2).

The validation suite of Section 4 thus shows Milstein at a small disadvantage for European pricing. Its advantage materialises in pathwise applications. We mention it here because confronting the irrelevance of Milstein for vanilla European pricing is part of internalising the strong/weak distinction.

4 Empirical validation strategy

The validation is the experimentally most informative part of the algorithm. We design a single comparison study that exhibits both the strong-order improvement and the weak-order tie of Milstein relative to Euler.

4.1 Test 1: side-by-side strong order

The most distinctive part of the validation script is a comparison of the two schemes' strong errors on the *same* grid of step counts and using the *same* Brownian paths. The protocol:

1. Choose a logarithmic grid of step counts N_k matching the strong-order grid of the Euler–Maruyama pricer, $N \in \{10, 30, 100, 300, 1000, 3000\}$.
2. For each N_k, sample M matrices of Brownian increments ΔW of shape (M, N_k).
3. Run the Euler scheme on these increments to obtain $\hat{S}_T^{\text{Euler},(i)}$ and the Milstein scheme on the *same* increments to obtain $\hat{S}_T^{\text{Milstein},(i)}$.
4. Compute the exact terminal $S_T^{(i)}$ from $W_T^{(i)} = \sum_n \Delta W_n^{(i)}$ via the closed-form GBM solution.
5. Estimate $\text{strong}_{\text{scheme}}(h_k) \approx M^{-1} \sum_i |S_T^{(i)} - \hat{S}_T^{\text{scheme},(i)}|$ for both schemes.
6. Plot both error curves on a single log-log axis.

The expected result is two parallel lines on log-log: Euler with slope $\frac{1}{2}$, Milstein with slope 1. Beyond confirming the two convergence orders, the side-by-side plot makes the asymmetry visually obvious: the ratio of Euler's strong error to Milstein's grows as $h^{-1/2}$, by a factor $\sqrt{h_{\text{largest}}/h_{\text{smallest}}} = \sqrt{300} \approx 17$ across the grid. In the recorded run of Test 1 of `validate_mc_european_milstein.py` the ratio is 8.6 at $h = 0.1$ and 145 at $h = 3.3 \times 10^{-4}$.

4.2 Test 2: weak order under common random numbers

A direct adaptation of the Euler–Maruyama weak test, applied to Milstein. The expected slope is 1, the same as Euler. This test serves as the empirical confirmation that Milstein *does not* improve the European pricer: same weak order, same asymptotic quality.

The grid restriction discussed in the Euler–Maruyama note applies here: the bias must dominate the residual MC noise of the CRN-corrected difference. With $M = 200{,}000$ and $N \in \{10, 20, 50, 100\}$, Milstein resolves the slope cleanly: because it tracks the exact path closely, its CRN difference has a standard deviation of about 0.008 at $N = 100$, against 0.24 for Euler, whose finest points sit near the noise floor.

4.3 Tests 3 and 4: coherence and validation

Identical in structure to the corresponding tests of the Euler–Maruyama pricer, applied to the Milstein pricer. No conceptual novelty.

5 Algorithmic specification

Algorithm 1 Milstein Monte Carlo for the European call (GBM)

Require: Spot S_0, strike K, maturity T, rate r, volatility σ, step count N, sample size M, confidence level $1-\beta$, random generator `rng`.
Ensure: Estimate $\hat{C}$, half-width h_M, sample variance S_M^2.

$h \leftarrow T/N, \quad D \leftarrow e^{-rT}$
for $i = 1, \dots, M$ **do**
  $S^{(i)} \leftarrow S_0$
  for $n = 0, 1, \dots, N-1$ **do**
    Draw $\Delta W \sim \mathcal{N}(0, h)$ using `rng`
    $S^{(i)} \leftarrow S^{(i)} \cdot \left(1 + rh + \sigma \Delta W + \frac{1}{2}\sigma^2((\Delta W)^2 - h)\right)$
  end for
  $Y_i \leftarrow D \cdot \max(S^{(i)} - K, 0)$
end for
$\hat{C} \leftarrow M^{-1} \sum_i Y_i$
$S_M^2 \leftarrow (M-1)^{-1} \sum_i (Y_i - \hat{C})^2$
$h_M \leftarrow z_{\beta/2} \sqrt{S_M^2 / M}$
return $(\hat{C},\ h_M,\ S_M^2)$

The implementation vectorises the inner loop across paths (advance all M states simultaneously) using `numpy.cumprod` on the matrix of factors, identical in structure to the Euler–Maruyama implementation. The only changes to the code are the addition of the quadratic corrector term in the factor and the corresponding docstrings; the two path samplers in `quantlib.gbm` are of the same length, about 50 lines each.

6 Bibliographic notes

The Milstein scheme is from Milstein [4] and is treated in detail in Kloeden & Platen [3], Section 10.3. The empirical side-by-side comparison study designed here is a direct adaptation of the comparison study in Higham [2], Section 5. For the practical relevance of Milstein in finance applications, Glasserman [1, Section 6.1] provides the context-specific discussion: when does the strong order of a scheme matter for the financial applications at hand.

References

[1] P. Glasserman, *Monte Carlo Methods in Financial Engineering*, Springer, 2004.

[2] D. J. Higham, *An algorithmic introduction to numerical simulation of stochastic differential equations*, SIAM Review, 43(3):525–546, 2001.

[3] P. E. Kloeden and E. Platen, *Numerical Solution of Stochastic Differential Equations*, Springer, 1992.

[4] G. N. Mil'shtein, *Approximate integration of stochastic differential equations*, Theory of Probability and its Applications, 19(3):557–562, 1974.